

Unit 10, Lesson 3: Creating Displays from Survey Data Plots

Benchmarks

MA.6.DP.1.1 Recognize and formulate a statistical question that would generate numerical data.

MA.6.DP.1.3 Given a box plot within a real-world context, determine the minimum, the lower quartile, the median, the upper quartile, and the maximum. Use this summary of the data to describe the spread and distribution of the data.

MA.6.DP.1.4 Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers, and the range.

MA.6.DP.1.5 Create box plots and histograms to represent sets of numerical data within real-world contexts.

Learning Targets

- ☐ I can create a statistical question and collect survey data.
- ☐ I can determine ways to collect truthful data.
- ☐ I can create box plots and histograms to display data.
- ☐ I can describe and interpret data displays representing a real-world context.

10.3.1 Warm-Up: Creating Statistical Questions

Create three statistical questions that could be used in a class survey to collect quantitative data.

1.

2.

3.

10.3.2 Collaboration: Determining Ways to Collect Truthful Data

Form a group of 3-4 classmates and discuss the questions everyone in your group created in the Warm-Up.

- 1. Choose the best three statistical questions that will produce quantitative data.

Statistical Question	Ways to Collect Truthful Data
1.	
2.	
3.	

- 2. With your group, discuss ways to collect truthful data for each of the questions. Summarize your discussion in the table for each statistical question.
- 3. Explain what makes these good statistical questions.

10.3.3 Activity: Class Survey

Data Collection Directions:

1. As a group, choose one question to survey the class.
2. Write your group's question in the top of the data chart.
3. Place the data chart in a location easily accessible by your classmates. Your teacher may have further instructions.
4. Follow your teacher's directions to answer each group's question.

My Group's Statistical Question:			
Name	Answer	Name	Answer
1.		15.	
2.		16.	
3.		17.	
4.		18.	
5.		19.	
6.		20.	
7.		21.	
8.		22.	
9.		23.	
10.		24.	
11.		25.	
12.		26.	
13.		27.	
14.		28.	

Once the results are collected from the whole class, work with your group to create a presentation displaying the data from your survey results.

Data Display Directions:

- Present the data in a neat, organized table.
- Find the five-number summary and create a box plot.
- Create a frequency table and a histogram.
- Describe the distributions shape, center, spread, and unusual features.
- Summarize the class' results using statistical terminology.

10.3.4 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each topic.



I need help.



I got it.



I could help someone.

Determining ways to collect truthful data			
Finding a five-number summary			
Creating a box plot			
Creating a frequency table			
Creating a histogram			
Describing data displayed on a box plot, frequency table, and histogram			